MINI PROJECT REPORT

Phishing Simulation Tracker

A MySQL database for campaign tracking, incident logging and automated risk scoring

Database Management Systems

Project team

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Abstract

The Phishing Simulation Tracker is a MySQL database designed to model employee participation in controlled phishing-awareness campaigns. It stores employee and campaign information, records link clicks and credential submissions, and updates an employee risk score when credentials are submitted. A database trigger adds 50 points for each first-time credential submission recorded on a simulation log and changes the account status to “LOCKED - Breach Detected” when the score reaches 100. Stored procedures guide the simulation sequence, provide a SecOps dashboard, and reset an employee after remediation. A forensic view joins the three core tables to support incident review. The project demonstrates relational keys and constraints, triggers, procedures, conditional logic, joins and views.

Project at a glance

Item	Description
Database	MySQL database named cyber_sim
Core entities	employee, campaign and simulation_log
Automation	One AFTER UPDATE trigger and four stored procedures
Audit	forensic_audit_log view
Risk rule	+50 on each false-to-true credential submission; lock at 100 or more

1. Introduction

Phishing simulations help organizations assess how employees respond to realistic email and web lures without relying on real attacks. This project provides the data layer for such an exercise. It maintains a roster of employees, describes campaign payloads, and records whether each employee clicked a simulated link or submitted credentials.

The database is organized so that one employee may participate in multiple campaigns and each campaign may target multiple employees. The simulation_log table resolves this many-to-many relationship and stores the response for each employee-campaign pair. A unique constraint prevents duplicate log rows for the same pair.

Objectives

- Maintain employee roles, risk scores and account status.
- Define phishing campaigns and their payload types.
- Record link clicks and credential submissions in sequence.
- Apply risk scoring and account lockout automatically.
- Provide a queryable incident view and a SecOps summary.

2. ER diagram

The diagram shows simulation_log as the associative entity between employee and campaign. Each log row references exactly one employee and one campaign; either parent may be associated with zero or many log rows.

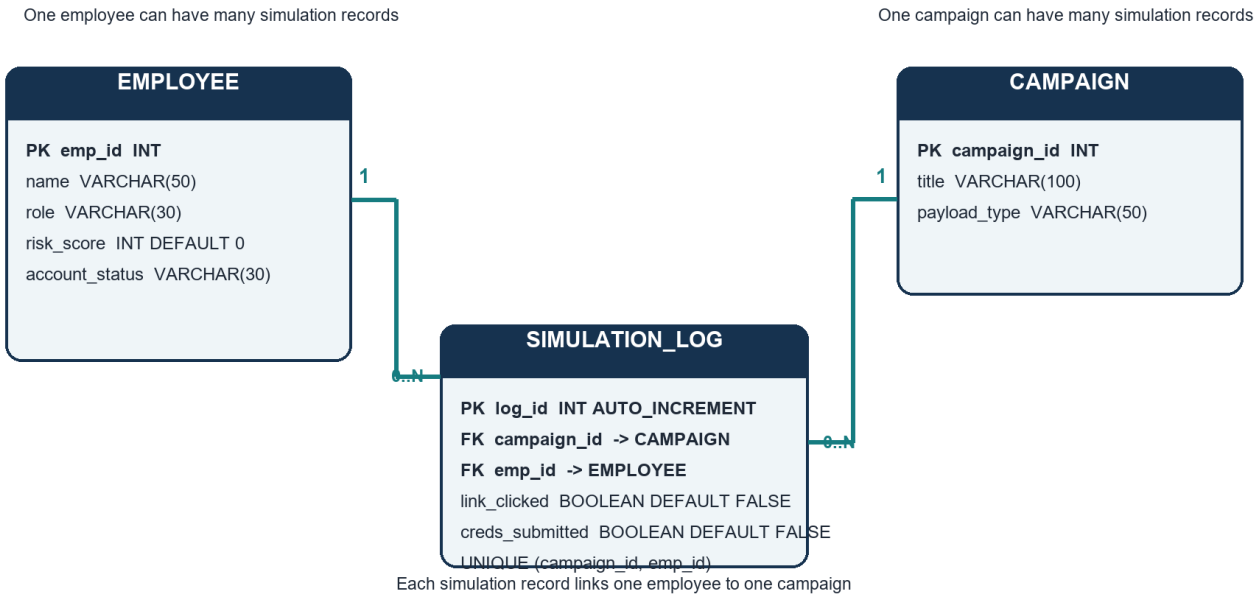


Figure 1. Entity relationship diagram derived from the supplied MySQL schema.

Relationship notes

EMPLOYEE (1) to SIMULATION_LOG (0..N) and CAMPAIGN (1) to SIMULATION_LOG (0..N). Together, the two one-to-many relationships implement an employee-to-campaign many-to-many relationship. The unique key on (campaign_id, emp_id) limits the design to one tracking row per employee per campaign.

3. Table schema

The schema below reflects the column types, defaults, keys and constraints in the supplied SQL source.

EMPLOYEE

Column	Type	Constraint / default
emp_id	INT	Primary key
name	VARCHAR(50)	
role	VARCHAR(30)	
risk_score	INT	DEFAULT 0
account_status	VARCHAR(30)	DEFAULT ACTIVE

CAMPAIGN

Column	Type	Constraint / default
campaign_id	INT	Primary key
title	VARCHAR(100)	
payload_type	VARCHAR(50)	

SIMULATION_LOG

Column	Type	Constraint / default
log_id	INT	Primary key; AUTO_INCREMENT
campaign_id	INT	Foreign key to campaign.campaign_id
emp_id	INT	Foreign key to employee.emp_id
link_clicked	BOOLEAN	DEFAULT FALSE
creds_submitted	BOOLEAN	DEFAULT FALSE
(campaign_id, emp_id)	Composite UNIQUE	One log per employee per campaign

Seed data

The SQL inserts ten employees (IDs 101–110) and five campaigns (IDs 1–5). Employee risk scores and statuses use their defaults when inserted. Campaign examples include “Fake IT Update,” “Payroll Document,” “Free Coffee Promo,” “Urgent CEO Request,” and “Password Expiry,” with payload types including credential harvester, malicious attachment, link click tracker and spear phishing.

4. Application and database logic

The application is implemented in MySQL through stored procedures, a trigger and a view. The procedures expose the main simulation and incident-response actions while the trigger enforces the risk rule when a credential submission is recorded.

Component	Purpose and behavior
click_link(target_emp, camp_id)	Creates a log row if needed using INSERT IGNORE, marks the link as clicked and returns a status message.
submit_creds(target_emp, camp_id)	Checks that the employee clicked the campaign link first. If so, marks credentials submitted; otherwise returns a sequence error.
security_monitor trigger	AFTER UPDATE on simulation_log. Adds 50 to risk_score only when creds_submitted changes from FALSE to TRUE. Sets account_status to LOCKED - Breach Detected at a score of 100 or more.
secops_dashboard()	Lists employee, role, risk score and account status in descending risk-score order.
neutralize_threat(target_emp)	Resets risk score to 0, sets account status to active, and deletes the employee's simulation log records.
forensic_audit_log view	Joins the three tables, includes clicked records, aliases response fields for audit reporting and sorts newest log first.

Typical workflow

- Select an employee and campaign.
- Call click_link to create or update the response record.
- Call submit_creds only after the link click. The trigger applies the risk update.
- Review risk and account state using secops_dashboard().
- Use forensic_audit_log to inspect clicked incidents.
- After response and remediation, call neutralize_threat for the employee.

Implementation note

The lockout condition is evaluated after each first-time credential submission. At 50 points per submission, two successful submissions across different campaigns bring an employee to 100 and trigger the locked status. Repeating an update while creds_submitted is already TRUE does not add another 50 points because the trigger checks the old and new values.

5. Future improvement

The presentation identifies several extensions that could make the tracker more useful in a larger organizational setting:

- Training plans for employees with high risk scores.
- A campaign performance dashboard with click and credential-submission rates.
- Automated email campaign integration.
- Role-based access for SecOps users.
- Historical campaign analysis across multiple exercises.

Additional implementation work could also add input validation, timestamps for every action, explicit transaction handling, and test cases for boundary conditions such as exactly 100 risk points and repeated procedure calls. These are proposed extensions and are not part of the supplied SQL implementation.

6. Conclusion

The Phishing Simulation Tracker models the lifecycle of a phishing-awareness exercise using a compact relational schema. Its foreign keys and unique constraint preserve the employee-campaign relationships, while procedures guide the event sequence and the trigger applies risk scoring and lockout rules. The forensic view and SecOps dashboard provide basic reporting and incident review. The project demonstrates core DBMS concepts and provides a foundation for future training, analytics and campaign integration.

7. References

- Project SQL source: dbms_pro.txt, supplied with the project.
- Project presentation: Phishing_Simulation_Tracker_Updated.pptx, supplied with the project.
- MySQL 8.0 Reference Manual, CREATE TABLE, CREATE TRIGGER, CREATE PROCEDURE and CREATE VIEW statements: https://dev.mysql.com/doc/refman/8.0/en/ .

The SQL and presentation are the primary sources for the schema, sample data, risk rule and listed future scope. The MySQL manual is included as a syntax reference.

8. Screenshot of execution

The following screenshots document the supplied MySQL execution: a click and credential submission, the SecOps risk dashboard, and the forensic audit view.

```
mysql> call click_link(101,1);
+-----------------------------------------------------------------------------------------------------------------------+
| terminal_output                                                                                                       |
+-----------------------------------------------------------------------------------------------------------------------+
|
>>> TARGET ACQUIRED: employee 101
>>> Connection established to payload...
>>> Awaiting credential input...
 |
+-----------------------------------------------------------------------------------------------------------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> call submit_creds(101,1);
+-----------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------+
| terminal_output
                             |
+-----------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------+
|
>>> CREDENTIALS CAPTURED for employee 101
>>> Target fell for the simulation.
>>> Auto-updating organizational risk score...
>>> Current account status: ACTIVE
 |
+-----------------------------------------------------------------------------------------------------------------------------------------------------------
------------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

Figure 2. MySQL execution of click_link(101, 1) followed by submit_creds(101, 1).

```
mysql> call secops_dashboard();
+-----------+------------------+---------------------+--------------------------+
| employee  | department       | vulnerability_score | network_access           |
+-----------+------------------+---------------------+--------------------------+
| Lewis     | Sys Admin        |                 100 | LOCKED - Breach Detected |
| Ayrton    | Sales Rep        |                 100 | LOCKED - Breach Detected |
| Max       | Developer        |                  50 | ACTIVE                   |
| Michael   | HR Manager       |                  50 | ACTIVE                   |
| Niki      | Marketing Exec   |                  50 | ACTIVE                   |
| Fernando  | Data Analyst     |                  50 | ACTIVE                   |
| Kimi      | Intern           |                  50 | ACTIVE                   |
| Sebastian | Finance Lead     |                   0 | ACTIVE                   |
| Charles   | Customer Support |                   0 | ACTIVE                   |
| Alain     | Project Manager  |                   0 | ACTIVE                   |
+-----------+------------------+---------------------+--------------------------+
10 rows in set (0.00 sec)

Query OK, 0 rows affected (0.03 sec)
```

Figure 3. SecOps dashboard showing employee risk scores and account statuses.

```
mysql> select * from forensic_audit_log;
+-------------+-------------+-----------------+-------------------+---------------------+-----------------+-----------------+
| incident_id | target_name | department      | attack_vector     | payload             | link_compromised | data_exfiltrated |
+-------------+-------------+-----------------+-------------------+---------------------+-----------------+-----------------+
|          10 | Alain       | Project Manager | Payroll Document  | Malicious Attachment |               1 |               0 |
|           9 | Kimi        | Intern          | Urgent CEO Request | Spear Phishing      |               1 |               1 |
|           8 | Fernando    | Data Analyst    | Password Expiry   | Credential Harvester |               1 |               1 |
|           7 | Ayrton      | Sales Rep       | Free Coffee Promo | Link Click Tracker  |               1 |               1 |
|           6 | Ayrton      | Sales Rep       | Payroll Document  | Malicious Attachment |               1 |               1 |
|           5 | Niki        | Marketing Exec  | Fake IT Update    | Credential Harvester |               1 |               1 |
|           4 | Michael     | HR Manager      | Password Expiry   | Credential Harvester |               1 |               1 |
|           3 | Lewis       | Sys Admin       | Urgent CEO Request | Spear Phishing      |               1 |               1 |
|           2 | Lewis       | Sys Admin       | Free Coffee Promo | Link Click Tracker  |               1 |               1 |
|           1 | Max         | Developer       | Fake IT Update    | Credential Harvester |               1 |               1 |
+-------------+-------------+-----------------+-------------------+---------------------+-----------------+-----------------+
10 rows in set (0.00 sec)
```

Figure 4. forensic_audit_log query showing campaign incidents and recorded responses.